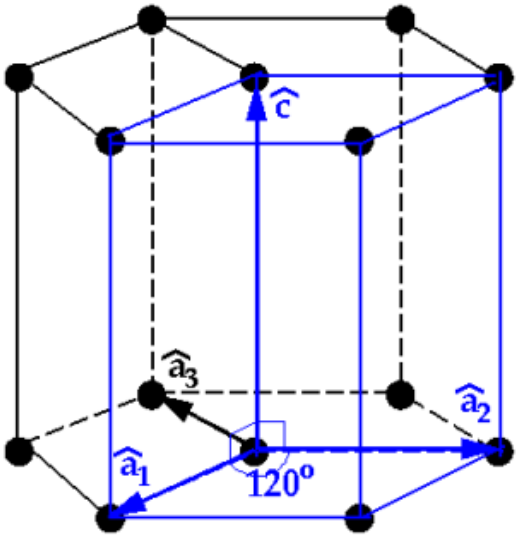
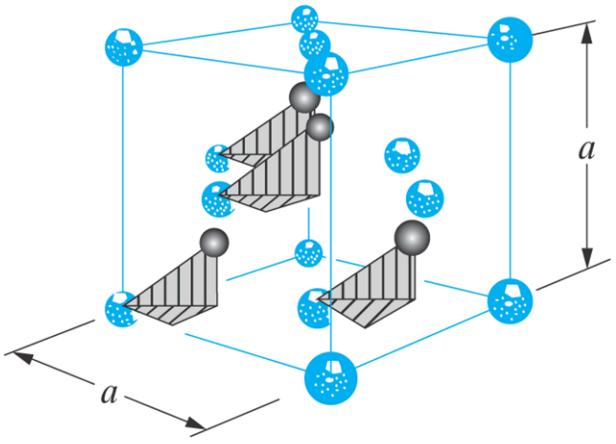
EE2021: Tutorial -3 (August 28th, 2025)

1. Derive the packing density of BCC and FCC cubic lattice

2. A hexagonal prism is shown. Find the Miller indices of all six side walls using a_1, a_2 and c as the three coordinate axes as shown below. Ignore a_3



3. If a=5.43 Å, find the density of atoms in Silicon lattice



4. Write the primitive unit vectors in real space and then find the primitive unit vectors in reciprocal space for this 2-D lattice. Assume $\gamma=120^0$

